DISASTER RESOURCE COORDINATION PLATFORM

CS543 – Web Services Project Submission

Group ID: 02

Team Members

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1. Project Overview

The Disaster Resource Coordination Platform is a web-based system designed to support disaster response, shelter management, field reporting, and resource coordination.

The platform helps shelter staff report resource availability and field conditions, enables coordinators to monitor shelter status, and demonstrates web-services concepts through API communication and service contracts.

2. Objectives

• Support structured reporting and coordination for disaster-response activities.

• Demonstrate HTTP APIs, service-oriented architecture, and API documentation.

• Apply input validation, consistent error responses, and automated testing.

• Maintain implementation artifacts and documentation in a version-controlled repository.

3. Technology and Interfaces

Backend: Node.js and Express.js for the reports API and HTTP handling.

API contract: OpenAPI 3.0.3 documentation.

Testing: automated server-side tests configured in the server package.

Additional data stores, frontend components, SOAP/WSDL interfaces, and integrations should be verified using the repository's implementation and assignment documents before claiming they are operational.

4. Assignment 01–06 Documentation

Assignment 01: Requirements and project/domain documentation.

Assignment 02: Design and data-flow documentation included in the project.

Assignment 03: Integration and service communication artifacts included in the project.

Assignment 04: Disaster Resource Coordination design/DFD documentation.

Assignment 05: API/service contract work; refer to the repository specifications for exact endpoints and behavior.

Assignment 06: Reports API improvements covering structured errors, content negotiation, rate limiting, health-check failure handling, validation responses, creation Location headers, OpenAPI documentation, and automated tests.

5. API Reliability and Error Handling

The reports API implementation includes structured error responses and field-level validation details. It supports JSON/XML content negotiation and returns HTTP 406 for unsupported Accept formats. Rate-limited requests use HTTP 429 with Retry-After. Health-check database failures are represented as HTTP 503 with Retry-After. Validation failures use HTTP 422. Creation/replay responses include a Location header where implemented.

The recorded server test run passed 16 of 16 tests. This result refers to the local test run; it does not establish that a remote deployment has been tested.

6. Repository Contents

The repository contains server application code, reports API modules, OpenAPI documentation, tests, notes, and design/integration artifacts. Refer to the GitHub repository for the current folder structure and implementation.

7. GitHub Repository

URL: https://github.com/amankumar242323-png/disaster-resource-coordination

Branch: main

8. Demonstration Video / Google Drive

Google Drive demonstration link: [REPLACE THIS WITH YOUR REAL GOOGLE DRIVE LINK]

Before final submission, replace this placeholder with the actual demonstration URL and set sharing to “Anyone with the link” with appropriate viewing permission.

9. How to Review and Test

1. Open the GitHub repository and review the assignment artifacts and API documentation.

2. Follow the setup instructions in the server package and repository documentation.

3. Install dependencies and run the server tests using the documented project commands.

4. Start the service and test its endpoints according to the API documentation.

10. Conclusion

This submission organizes the Disaster Resource Coordination project implementation and documentation for CS543 Assignments 01–06. The repository is the source of truth for the code and artifacts. Verify the assignment checklist and add the real Google Drive demonstration link before submitting.